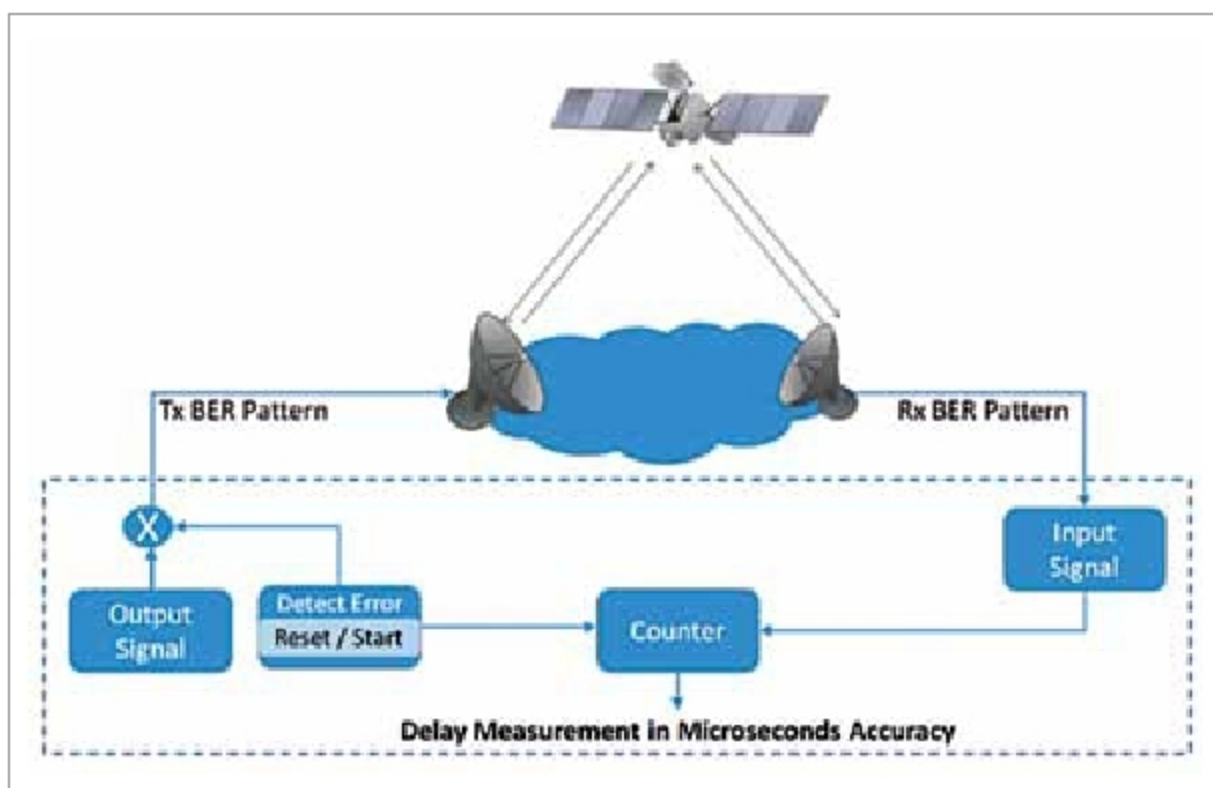




Satellite communication testing (Credit: www.gl.com)

when these are performed under an environment that is close to reality. Thus, GNSS simulation must include proper simulation of satellite orbits, interfering signals, signal propagation characteristics, characteristics of receiver antenna, and receiver environment.

Ground station and ground terminal testing. Once a satellite is on its target orbit, several routines need to be performed to ensure optimum performance of the entire chain of satellite communications. Validation, monitoring, and performance test of ground stations are of vital importance for the satellite operator.

Vibration testing. Vibration tests are very costly to carry out and are among the most crucial checks to conduct. The collected data from the vibration test allows the analysis of the satellite structure design to meet the vibration requirements. Vibration test is generally done before the thermo-vacuum test by applying various loads on the launch vehicle as follows:

- Static loading tests the vibrations due to accelerations experienced during flight
- Harmonic loading simulates the vibrations that the satellite will experience during launch
- Random loading tests random vibrations in time and frequency domain
- Shock testing tests for mechan-

ical shocks during the liftoff and subsequent launch trajectory

- Acoustic test simulates a vibration environment to verify whether the satellite can function normally after bearing acoustic vibrations

Satellite payload testing. Satellite payload systems must operate reliably in space, providing reliable service for decades of operational life. Additional tests will include measurements to learn each satellite's exact centre of gravity and mass to ensure its compatibility with its launch vehicle. It also helps to control the orientation when the satellites are in orbit, which can determine how long they will operate in space.

Electromagnetic compatibility. Satellites also need to get tested for shielding against external radio signals. The electromagnetic (EM) interference and compatibility tests are used to understand EM field response produced by the satellite itself under normal operation and to determine the EM interference limit that satellites can tolerate.

Hardware-in-the-loop simulation (HLIS) test. A HILS facility is used to test one of the most complex systems onboard, the Attitude and Orbit Control System (AOCS) of satellites in a closed-loop, catering to the stringent requirements of the spacecraft mission. During HILS tests, deficiencies in the spacecraft hardware, software, interfaces, etc, are determined and corrected to achieve flawless opera-

tion of AOCS in orbit.

Subsystem testing. It includes battery voltage and bus voltage testing, primary and secondary battery voltage testing, solar panel voltage checking, actuator and sensors testing, momentum wheel testing, magnetic voltage, sun sensor testing, indoor and outdoor testing, RF emission testing, pressure checking, fuel tank and oxidiser tank testing, solenoid valve checking, static and dynamic load testing, and satellite thrusters testing.

Some satellite testing solutions providers

For many years, only governments were involved in the construction and testing of satellites. However, these days large and small private companies are increasingly investing in satellites for commercial purposes like television and radio signals, telecommunications, and even military applications.

NTS provides a network of facilities to conduct the necessary tests, including building gigantic climate chambers, to analyse how a satellite would respond to the vacuum of space or to extreme changes in temperature. It also conducts tests for vibration, solar radiation, dealing with the dust of space or pyroshock, which might occur during the booster separation stage or the satellite separation stage from explosive bolts.

NASA and some other private companies are exploring the ways of satellite servicing, which can make spaceflight more sustainable, affordable, and resilient. It is testing space robots to diagnose, maintain, and extend a spacecraft's life.

GNSS test solutions from Rohde & Schwarz and some others enable testing subsystems, assemblies, and components to validate the performance of ground station terminals and in-orbit service under well-defined and controlled conditions. All GNSS scenarios are fully customisable and repeatable.

GL Communications provides various test solutions to monitor and test backhaul networks, no matter what type of media, bandwidth, or signalling is over the satellite network. **EFY**